Mathematics - Module 2 (Applied)

A.Y. 2023/24

Elementary Financial Calculus

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Contents

1. Elementary financial operations

The starting point of elementary financial calculus is the definition of an elementary financial operation.

Definition 1. (*Elementary financial operation*). An exchange between two amounts of money available on different dates is called an *elementary financial operation.*

We have two particular types of financial operations:

- *accumulation operations* (or accumulation problems)
- *discount operations* (or discount problems).

1.1. Accumulation operations

An elementary financial operation is called an accumulation operation when the target is to move an amount of money forward in time.

Definition 2. (*Accumulation operation*). Let M, C be positive real numbers. An *accumulation operation* is an exchange between an amount of money available today C (principal or invested capital) with another one M (finale value or accumulated value) available in the future (at a date $t \geq 0$).

We can describe an accumulation operation using either a table or a sequence of pairs as follows

Dates	0	t
Cash flows	$-C$	M

or $\{(0, -C), (t, M)\}$

therefore we have an outflow at 0 ($-C$) and an inflow at t (M). The *lifetime* is t and the difference between M and C, indicated with I, is said to be the interest generated by the accumulation operation, that is

$$I := M - C$$

The ratio $f(t)$ defined by

$$f(t) := \frac{M}{C} = \frac{\text{Final value of } C \text{ at } t}{\text{Principal at } 0}$$

is called the *accumulation factor* at $t \geq 0$. Its financial meaning is "the amount of money at t for each euro invested at 0". Given C and $f(t)$, the basic formula in order to calculate M for an accumulation problem is

$$M = C \cdot f(t) \qquad \text{for any } t \geq 0$$

that is the final value M is calculated by multiplying the principal C by the accumulation factor evaluated at t.

1.2. Discount operations

An elementary financial operation is called a discount operation when the target is to move an amount of money backward in time.

Definition 3. (*Discount operation*). Let S, A be positive real numbers. A *discount operation* is an exchange between an amount of money S (nominal value) available at $t \geq 0$ with another one A (present value or discounted value) available today (at $t = 0$). We can describe a discount operation using the following table

Maturities	0	t
Cash flows	A	$-S$

or $\{(0, A), (t, -S)\}$

therefore we have an outflow at t $(-S)$ and an inflow at 0 (A). The *lifetime* is t and the difference between S and A, indicated with D, is said to be the discount generated by the discount operation, that is

$$D := S - A$$

The ratio $\varphi(t)$ defined by

$$\varphi(t) := \frac{A}{S} = \frac{\text{Present value of } S \text{ at } 0}{\text{Nominal value at } t}$$

is called the *discount factor* at $t \geq 0$. Its financial meaning is "the amount of money at 0 for each euro available at t". Given S and $\phi(t)$, the basic formula in order to calculate A for a discount problem is

$$A = S \cdot \varphi(t) \qquad \text{for any } t \geq 0$$

that is the present value A is calculated by multiplying the nominal value S by the discount factor evaluated at t.

Example 4. The formula $f(2) = 1.15$ has the following financial meaning: "we obtain €1.15 in two years for €1 invested today". Meanwhile, the financial meaning of the formula $\varphi(7/12) = 0.94$ is "today we get €0.94 for €1 available at $t = 7/12$ (7 months)".

Remark 5. $f(t)$ and $\varphi(t)$ are called *financial factors*, which are functions of t (the lifetime of the financial operation). We can have situations where the financial factors depend on t and some parameters. We have: (a) $f(t, \alpha)$ where the accumulation factor depends on t and a parameter α (which regulates the growth of the principal over time); (b) $\varphi(t, \beta)$ the discount factor depends on t and a parameter β (which regulates the contraction of the nominal value over time).

Example 6. Consider the accumulation factor $f(t) = 1 + \alpha t$ with $\alpha > 0$, $t \geq 0$. In this case, the final value at t of €1 invested at 0 depends on t (the lifetime) and α (which regulates the growth of the principal with respect to the lifetime t). Giving a value to α, for example $\alpha = 0.12$, we get $f(t) = 1 + 0.12t$ and the final value of €1 depends only on t (the lifetime).

1.3. Conjugated financial factors, annual interest rate

We have a special relationship between financial factors $f(t)$ and $\varphi(t)$, with $t \geq 0$.

Definition 7. (*Conjugated financial factors*). Let $f(t)$ and $\varphi(t)$ be, respectively, an accumulation factor and a discount factor. The financial factors $f(t)$ and $\varphi(t)$ are said to be *conjugated financial factors* when

$$\forall t \geq 0,\ f(t) \cdot \varphi(t) = 1$$

In this case, the accumulation and discount operations are said to be *symmetrical* operations.

Given $f(t)$, $\varphi(t) = 1/f(t)$ is the discount factor conjugated to $f(t)$. Given $\varphi(t)$, $f(t) = 1/\varphi(t)$ is the accumulation factor conjugated to $\varphi(t)$.

Example 8. Consider a principal of €1200 and the accumulation factor $f(t) = 1 + 0.1t$, with $t \geq 0$. The final value M and the interest I one year later are

$$M = 1200 \cdot f(1) = 1200 \cdot (1 + 0.1 \cdot 1) = 1320 \quad \text{and} \quad I = 1320 - 1200 = 120$$

The conjugated discount factor $\varphi(t)$ to $f(t)$ is defined as $\varphi(t) = 1/f(t) = (1 + 0.1t)^{-1}$, for any $t \geq 0$. The present value A and the discount D of a nominal value of € 1500 available at $t = 1.5$ years (one year and six months) are

$$A = 1500 \cdot \varphi(1.5) = \frac{1500}{1 + 0.1 \cdot 1.5} \approx 1304.35 \quad \text{and} \quad D = 1500 - 1304.35 = 195.65$$

that is we lose €195.65 to have today the present value of €1500 available at $t = 1.5$.

From now on we will use a special parameter, the *annual interest rate*.

Definition 9. (*Annual interest rate*). The interest generated by €1 invested for the first year is indicated with the symbol i and is called the *annual interest rate*.

$$\begin{cases} f(1) = \text{final value of €1 invested today for one year} \\ i = f(1) - 1 = \text{annual interest rate} \end{cases}$$

Example 10. Let's consider the accumulation factor $f(t) = 1 + 0.15t$, with $t \geq 0$. The annual interest rate i is

$$i = f(1) - 1 = (1 + 0.15 \cdot 1) - 1 = 0.15 = 15\%$$

2. An axiomatic approach

2.1. Cauchy functional equations

Let us start with the most classic of all functional equations (that is equations in which the unknown is a function and not a number): the functional equation of A.-L. Cauchy (1789 - 1857). It asks to identify all the real functions such that the following property

$$\forall x, y \in \mathbb{R}, \quad f(x + y) = f(x) + f(y)$$

is satisfied.

Theorem 11. (A. L. Cauchy). Let f be a continuous function at least at one point $x_0 \in \mathbb{R}$. Then

$$\forall x, y \in \mathbb{R}, \quad f(x+y) = f(x) + f(y) \quad \Leftrightarrow \quad f(x) = ax,\, a \in \mathbb{R}$$

We omit the proof.

Now we consider real functions with two variables instead of one.
We can ask the following question: which functions f satisfy the property $f(x+y,z) = f(x,z)+f(y,z)$, for any $x, y, z \in \mathbb{R}$? The answer is an immediate consequence of what we have seen: the only real functions, continuous at least at one point, that satisfy this property have the form $f(x,z) = a(z) \cdot x$, with $a(z)$ an arbitrary function of z.

Now we consider another functional equation, that is $f(x+y) = f(x) \cdot f(y)$ with $x, y \in \mathbb{R}$. The target is to find a function such that the image of the sum is equal to product of the images.

Theorem 12. Let f be a strictly positive function, continuous at least at one point $x_0 \in \mathbb{R}$. Then

$$\forall x, y \in \mathbb{R}, \quad f(x+y) = f(x) \cdot f(y) \quad \Leftrightarrow \quad f(x) = e^{mx},\, m \in \mathbb{R}$$

Proof. ($\Rightarrow$) Assume that $\forall x, y \in \mathbb{R}, \quad f(x+y) = f(x) \cdot f(y)$. Taking the logarithm of each side we get

$$\ln f(x+y) = \ln \left[f(x) \cdot f(y) \right] \quad \Leftrightarrow \quad \ln f(x+y) = \ln f(x) + \ln f(y)$$

and, using the theorem 1.11, a linear function $f(x) = mx$ exists, with $m \in \mathbb{R}$, such that $\ln f(x) = mx$. Therefore $f(x) = e^{mx}$, $m \in \mathbb{R}$. ($\Leftarrow$) Assume that $f(x) = e^{mx}$, $m \in \mathbb{R}$. Let x, y be real numbers. Starting from the first side we have

$$f(x+y) = e^{m(x+y)} = e^{mx+my} = e^{mx} \cdot e^{my} = f(x) \cdot f(y)$$

2.2. Capitalization axioms

The primitive concepts on which the basic capitalization model is based are: (1) the amount of money or capital C available today (at $t = 0$); (2) the lifetime t of the capitalization; (3) the amount of money or final value M available in a future date (at $T > 0$). The following axioms indicate the relationships between the variables C, t, M ($C \geq 0$ and $t \geq 0$):

- **Axiom 1 (A1)**: the final value M depends on C and t, therefore we can write

$$M = M(C, t)$$

- **Axiom 2 (A2)**: let C_1, C_2 be two capitals, we have

$$\forall C_1, C_2 \geq 0,\, M(C_1 + C_2, t) = M(C_1, t) + M(C_2, t)$$

- **Axiom 3 (A3):** let t_1, t_2 be two lifetimes, we have

$$\forall t_1, t_2 \geq 0,\, t_1 < t_2 \Rightarrow M(C, t_1) \leq M(C, t_2)$$

- **Axiom 4 (A4):** If the lifetime is $t = 0$ then $M(C, 0) = C$.

Remark 13. (1) Axiom A.1 simply defines which variables are taken into account by the capitalization model; (2) Axiom A.2 is the real cornerstone of elementary financial mathematics: the final value of the sum of two capitals, that is $C_1 + C_2$, is equal to the sum of the final value of C_1 and the final value of C_2 (with the same lifetime); (3) Axiom A.3 excludes that a capital can lose value over time. If we understand monetary value (nominal) as "value", there is no doubt that A.3 is fully realistic: if, on the other hand, we understand real value (purchasing power) as "value", it is undoubtedly unrealistic. In what follows it will therefore be appropriate to always think of the monetary value. (4) Axiom A.4 excludes the presence of other costs: a capitalization with lifetime 0 does not cause any change on the value of capital C.

Apart from A.1 which simply indicates the relevant variables and A.3 which, with the observation we made, is completely realistic, the other axioms (A.4 and especially A.2) may sometimes be violated in practice. We point out two immediate consequences of the axioms: (i) from A.2 and A.3 we deduce that

$$\forall C \geq 0,\ \forall t \geq 0,\ M(C,t) \geq M(C,0) = C$$

that is the capital C is never less than the final value M, for any lifetime $t \geq 0$.(ii) From A.1, A.2 and A.3, if $C_2 > C_1$, it follows that

$$\begin{aligned} M(C_2,t) &= M(C_1 + C_2 - C_1, t) = M(C_1,t) + M(C_2 - C_1, t) \\ &\geq M(C_1,t) + M(C_2 - C_1, 0) = M(C_1,t) + C_2 - C_1 > M(C_1,t) \end{aligned}$$

that is, in short, $C_2 > C_1 \Rightarrow M(C_2,t) > M(C_1,t)$: the final value is a strictly increasing function of the capital C.
From the four axioms we can characterize the final value M as a product of the amount invested C and a function f, which depends on the lifetime t, as we can see in the following theorem.

Theorem 14. Let $C \geq 0$, $t \geq 0$ be, respectively, a capital and lifetime. All functions $M(C,t)$ which satisfy A1-A4 axioms are $M(C,t) = Cf(t)$ with the following properties: (1) $f(0) = 1$; (2) f is increasing.

Proof. Axiom A.2 is the Cauchy functional equation (with respect to two variables); therefore, according to the remark we have seen after Theorem 11, we can conclude that $M(C,t) = Cf(t)$; moreover, f must be increasing to satisfy Axiom A.3 and it has to be $f(0) = 1$ to satisfy Axiom A.4.

Example 15. Consider $M(C,t) = C(1 + at)$, with $C \geq 0, t \geq 0$ and $a > 0$ (Axiom A.1 is satisfied, in fact M depends only on C and t). Axiom A.2 is true, for any pair of capitals $C_1, C_2 \geq 0$, we have

$$M(C_1 + C_2, t) = (C_1 + C_2)(1 + at) = C_1(1 + at) + C_2(1 + at) = M(C_1,t) + M(C_2,t)$$

Finally, we observe that $f(t) = 1 + at$ satisfies the two properties required from Theorem 1.14: f is increasing (Axiom A.3) and $f(0) = 1$ (Axiom A.4).

Example 16. Consider $M(C,t) = C\left(1 + \int_0^t a(s)\,\mathrm{d}s\right)$, with $a(s)$ a positive differentiable function on $[0, +\infty)$. (Axiom A.1 is satisfied, in fact M depends only on C and t). Axiom A.2 is true, for any pair of capitals $C_1, C_2 \geq 0$, we have

$$M(C_1 + C_2, t) = (C_1 + C_2)\left(1 + \int_0^t a(s)\,\mathrm{d}s\right) = M(C_1,t) + M(C_2,t)$$

Finally, we observe that $f(t) = \left(1 + \int_0^t a(s)\,\mathrm{d}s\right)$ satisfies the two properties required from Theorem 1.14: f is increasing (Axiom A.3 is satisfied, from $f'(t) = a(t) \geq 0$ for any $t \geq 0$) and $f(0) = 1$ (Axiom A.4). Taking $a(s) = a$, with $a \geq 0$, we have

$$M(C,t) = C\left(1 + \int_0^t a(s)\,\mathrm{d}s\right) = C\left(1 + \int_0^t a\,\mathrm{d}s\right) = C\,(1 + at)$$

Remark 17. The proof of Theorem 1.14 highlights the consequences of each of the axioms: Axiom A.2 determines the structure of the final value: $M(C,t) = Cf(t)$ (i.e. as a product of the capital C and a function f evaluated at the lifetime t); Axiom A.3 leads to the increasing property of f and Axiom A.4 to $f(0) = 1$.
It is then clear what happens if axioms A.3 and A.4 are abandoned or modified. For example, replacing A.3 with the following:

$$\textbf{A.3*} \qquad \forall t_1, t_2 \geq 0,\, t_1 < t_2 \Rightarrow M\left(C, t_1\right) < M\left(C, t_2\right)$$

we would have that f has to be strictly increasing; and replacing A.3 with the following:

$$\textbf{A.3}^{\circ} \qquad \forall t \geq 0, M(C,t) \geq 0$$

we would have that it has to be $f(t) \geq 0$ for any $t \geq 0$ (and f is not necessarily increasing anymore).

It is clear that, from now on, it is sufficient to know the analytical expression of $f(t)$, that is it is sufficient to take $C = 1$: $f(t) = M(1,t)$. The problem of determining M is in fact solved as soon as $f(t)$ has been specified; the amount of any capital C on date t is in fact obtained by multiplying $f(t)$ by the amount of the capital C (the final value M is proportional to the amount of capital C invested today). The function $f(t)$ therefore has the meaning of final value on t of a unitary capital invested at date 0.

Definition 18. (*Set of financial laws*). Let f be an accumulation factor defined for any $t \in [0, +\infty)$ with f increasing and $f(0) = 1$. A family of accumulation factors $\{f(t,\alpha)\}$, depending on a parameter α which varies in a given set A, is called a set of financial laws.

Example 19. Consider $f(t) = (1+a)^t$, with $t \geq 0$ and parameter $a > 0$. We can verify that: (1) f is increasing on $[0, +\infty)$; (2) $f(0) = 1$. Therefore f is an accumulation factor and the set defined as follows

$$\left\{(1+a)^t, t \geq 0 \text{ and } a \in (0, +\infty)\right\}$$

is a set of financial laws. Fixing a specific value of a, the intertemporal exchange ratio will depend only on the lifetime $t \geq 0$.

Remark 20. We present, in the next section, some specifications of the functional form of the growth factor $f(t) = M(1,t)$: the first two are widely used in practice and, indeed, substantially exhaust the possibilities of real applications. They will be obtained by making hypotheses on the time dynamics of f so that the assumptions underlying the capitalization laws obtained with them become clear. Both specifications assume that f is differentiable and the different assumptions concern the increment

$$\Delta f = f(s+h) - f(s)$$

relative to the variation from the date s to $s+h$, $s \geq 0$, $h > 0$.

3. Remarkable financial laws

We see here two different ways to move money over time: (1) simple interest; (2) compound interest.

3.1. Simple interest

We will now look at *simple accumulation* and *simple* (or *rational*) *discount*. First of all, we build up the simple accumulation and then, using the notion of conjugated financial factors, we find the simple (or rational) discount. We start with an assumption about the calculation of the interest on simple accumulation.

■ Simple accumulation

Assumption (*Simple interest*)**.** The interest, indicated with I, is a function of C, t and i defined by the formula $I = C \cdot i \cdot t$, with $C, t \geq 0$ and $i > 0$. Therefore the final value M on simple accumulation, remembering that $M = C + I$, is

$$M = C + I = C + C \cdot i \cdot t = C\,(1 + it) \tag{1.1}$$

and $f(t) = 1 + it$, with $t \geq 0$, is called the *simple accumulation factor*.

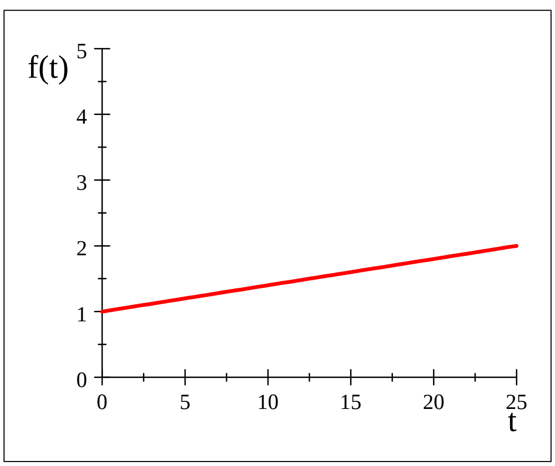

Figure 1

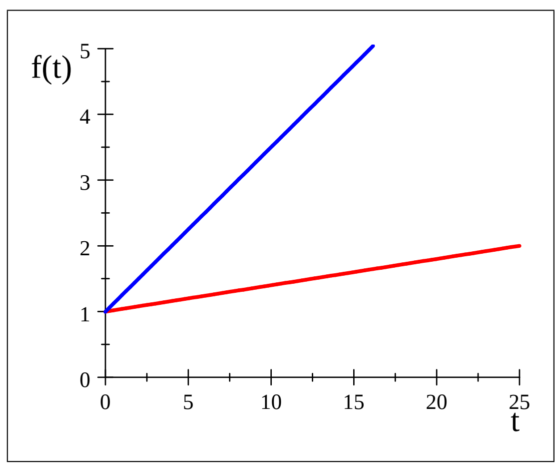

Figure 2

Figure 1 shows the graph of a simple accumulation factor with an annual interest rate of 4%. Figure 2 shows the graphs of simple accumulation factors with annual interest rates of 4% (red line) and 25% (blue line), where the key role of these parameters in governing capital growth over the time is clear (the two interest rates are the slopes of the two half-lines).

Example 21. Consider a principal of €1000, the simple annual interest rate $i = 0.12$ and $f(t) = 1 + 0.12t$, with $t \geq 0$. Calculate the final value M and interest I considering the following different lifetimes $t = 2$ (two years), $t = 3/12$ (three months) and $t = 125/365$ (125 days), as follows:

$$\begin{cases} M = 1000 \cdot f\,(2) = 1240 \\ I = 1240 - 1000 = 240 \end{cases} ; \quad \begin{cases} M = 1000 \cdot f\,(3/12) = 1030 \\ I = 1030 - 1000 = 30 \end{cases}$$

and $M = 1000 \cdot f\,(125/365) = 1041.1$ with $I = 1041.1 - 1000 = 41.1$.

Remark 22. We can find the simple interest accumulation factor as follows. Let f be a differentiable accumulation factor over $(0, +\infty)$. Suppose that the variation, indicated with Δf, when we move from s to $s+h$ (with $h \neq 0$) is only proportional to h as follows

$$f(s+h) - f(s) = a(s)h + o(h) \quad \text{as } h \longrightarrow 0,$$

where $a(s) \geq 0$ is a proportionality constant which depends on s, generally different from date to date. We have

$$\frac{f(s+h) - f(s)}{h} = a(s) + o(1) \qquad \Rightarrow \; f'(s) \; = a(s)$$

and integrating from 0 to t, and remembering that it must be $f(0) = 1$, we find

$$f(t) = 1 + \int_0^t a(s)\mathrm{d}s$$

In the simplest case, if $\forall s \geq 0$, $a(s) = a > 0$ (i.e. a constant function), we have $f(t) = 1 + at$.

■ **Simple (or rational) discount**

We start from the following two conditions: (1) the simple (or rational) discount is the conjugate operation of simple accumulation, therefore $f(t) \cdot \varphi(t) = 1$, with $t \geq 0$; (2) the simple accumulation factor is $f(t) = 1 + it$, with $t \geq 0$. The simple discount factor $\varphi(t)$ conjugated to $f(t)$ is

$$\varphi(t) = \frac{1}{f(t)} = \frac{1}{1+it} \qquad \text{for any } t \geq 0$$

and the present value A and the discount D of a nominal value S available on t using the simple discount are

$$A = S \cdot \frac{1}{1+it} \quad \text{and} \quad D = S - A = S - \frac{S}{1+it} = \frac{S \cdot i \cdot t}{1+it}$$

3.2. Compound interest

We will now look at *compound accumulation* and *compound discount*. First of all, we build up the compound accumulation and then, using the notion of conjugated financial factors, we obtain the compound discount.

■ **Compound accumulation**

We start with an assumption about the calculation of the interest during the accumulation problem. We consider the following accumulation problem: today we invest a principal C for a lifetime t. We assume that t is the sum of different periods t_1, t_2, ..., t_n, that is

$$t_1 + t_2 + \cdots + t_n = t \text{ (operation lifetime)}$$

Assumption (*Periodic interest capitalization*)**.** The interest calculated in each period is used to calculate the interest in the next one (i.e. the interest at the end of each period is added to the principal, used to calculate it, and this sum becomes the new principal for the next period). In each period we use simple accumulation with annual interest rate i. Indicating with M_s the final value on s $(s = 1, 2, ..., n)$, we get

- the final value at the end of the first period is

$$M_1 = C + C \cdot i \cdot t_1 = C\,(1 + it_1)$$

- the final value at the end of the second period (the interest of the second period is calculated from M_1 and not from C) is

$$M_2 = M_1 + M_1 \cdot i \cdot t_2 = M_1\,(1 + it_2) = C\,(1 + it_1)\,(1 + it_2)$$

- the final value at the end of the third period is

$$M_3 = M_2 + M_2 \cdot i \cdot t_3 = M_2\,(1 + it_3) = C\,(1 + it_1)\,(1 + it_2)\,(1 + it_3)$$

...

- the final value at the end of the n-th period is

$$M_n = M_{n-1} + M_{n-1} \cdot i \cdot t_n = M_{n-1}\,(1 + it_n) = C\,(1 + it_1)\,(1 + it_2) \cdots (1 + it_n)$$

The final value M on t using compound accumulation (and interest capitalization at the end of each period) is

$$M = C \prod_{s=1}^{n} (1 + i\,t_s) = C \cdot (1 + it_1) \cdot (1 + it_2) \cdot \dots \cdot (1 + it_n) \tag{1.2}$$

From now on, to calculate the final value M on compound accumulation, of a principal C invested in 0, at maturity t, with $t = n$ (the lenght of each period is equal to one), the following formula will be used

$$M = C \cdot (1 + i)^t \tag{1.4}$$

and $f\,(t) = (1 + i)^t$ is called the *compound accumulation factor* (where the interest capitalization occurs at the end of each year and at maturity). From now on, we will use the latter formula for any assigned maturity t (also when the maturity t is not an integer), as this is the use in practical applications.

Example 23. Consider a principal of € 2000, the compound annual interest rate $i = 0.05$ and $f(t) = 1.05^t$, with $t \geq 0$. Calculate the final value M and interest I considering different lifetimes t (remember that lifetime t is in years!):

- if t is two years ($t = 2$) then

$$M = 2000 \cdot f\,(2) = 2000 \cdot 1.05^2 = 2205 \quad \text{and} \quad I = 205$$

- if t is three months ($t = 3/12$) then

$$M = 2000 \cdot f\,(3/12) = 2000 \cdot (1 + 0.05)^{3/12} = 2024.54 \quad \text{and} \quad I = 24.54$$

- if t is 125 days ($t = 125/365$) then

$$M = 2000 \cdot f\,(125/365) = 2000 \cdot (1 + 0.05)^{125/365} = 2033.7 \quad \text{and} \quad I = 33.7$$

- if t is two years and seven months ($t = 2 + 7/12$) then

$$M = 2000 \cdot f\,(2 + 7/12) = 2000 \cdot (1 + 0.05)^{2+7/12} = 2268.66 \quad \text{and} \quad I = 268.66$$

Remark 24. The compound accumulation factor $f\,(t) = (1+i)^t$ is defined on $[0, +\infty)$ and has the following properties: (a) $f\,(0) = 1$ and $f\,(t) > 0$ for any $t \geq 0$; (b) $f\,(t)$ is strictly increasing on $[0, +\infty)$; (c) $f\,(t) \to +\infty$ as $t \to +\infty$ (f is divergent to $+\infty$ at $t \to +\infty$). The financial meaning of (b) is that the final value M increases as the lifetime t increases.

Figure 3 shows the graph of a compound accumulation factor with an annual interest rate of 4%. Figure 4 shows the graphs of compound accumulation factors with annual interest rates of 4% (red line) and 25% (blue line), where it is clear that different dynamics of growth of the capital depend only on the value of the annual interest rate used (clearly with the same maturity date and annual interest capitalization dates).

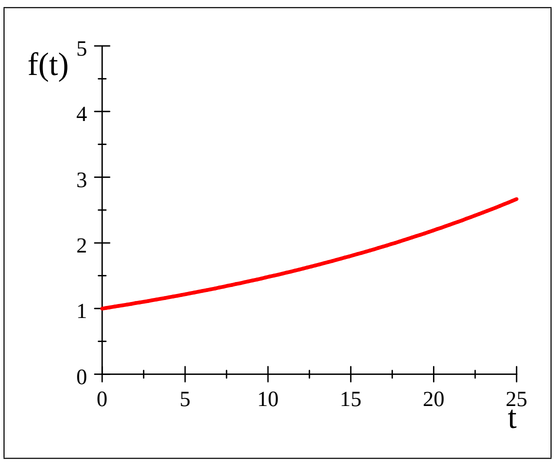

Figure 3

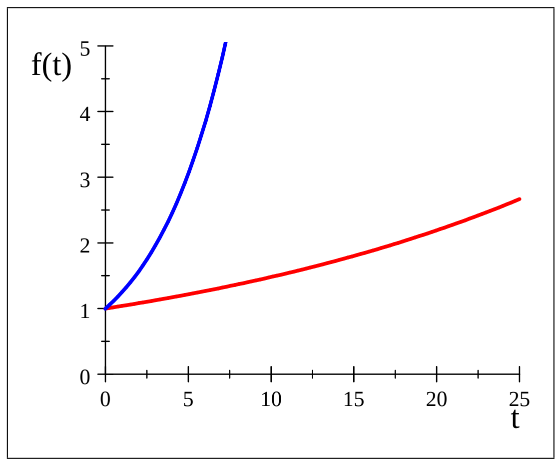

Figure 4

Remark 25. We can consider many dynamics of growth of capital C. For example, the accumulation factors $f : [0, +\infty) \to \mathbb{R}$ defined by the following formulas

$$\text{(a) } f(t) = \begin{cases} 1 + i_1\, t & \text{if} \quad t \leq 1 \\ \\ 1 + i_1 + i_2\,(t-1) & \text{if} \quad t > 1 \end{cases} \; ; \; \text{(b) } f(t) = \begin{cases} 1 + it & \text{if} \quad t \leq 1 \\ \\ (1+i)^t & \text{if} \quad t > 1 \end{cases}$$

give two different ways of evolution of a capital C (with $i_1, i_2 > 0$). Discuss case (a) with $i_1 = 0.04 < 0.05 = i_2$ and (b) using $i = 0.1$.

■ Compound discount

We start from the following two observations:

(a) compound discount is the inverse operation of compound accumulation, therefore they are symmetrical when $f\,(t) \cdot \varphi\,(t) = 1$, for any $t \geq 0$;

(b) the compound accumulation factor is $f\,(t) = (1+i)^t$, with $t \geq 0$.

The compound discount factor $\varphi\,(t)$, conjugated to $f(t)$, is defined by

$$\varphi(t) = \frac{1}{f(t)} = \frac{1}{(1+i)^t} = (1+i)^{-t} \qquad \text{for any } t \geq 0$$

The present value A and the discount D of a nominal value S available at t using the compound discount are

$$A = S \cdot (1+i)^{-t} \quad \text{and} \quad D = S - S \cdot (1+i)^{-t} = S \cdot \left[1 - (1+i)^{-t}\right]$$

Example 26. We have €10000 available three years from now and we want to calculate its present value using the compound discount with $i = 0.15$. We get

$$A = 10000 \cdot \varphi(3) = 10000 \cdot 1.15^{-3} \approx 6575.16$$

and the discount D associated is $D = S - A = 3424.84$.

■ Equivalent interest rates (in compound interest)

When we use compound accumulation, the interest capitalization dates are annual (as from the basic formula seen in (1.4)). But we can consider a compound accumulation where the interest capitalization dates are semiannual (or monthly, quarterly, daily). Indicating with i_m the interest rate if we want to consider the interest capitalization m times in a year (e.g. i_2 means that we have interest capitalization at the end of each semester, while i_{12} means that the interest capitalization will be at the end of each month), we have the following condition

$$\forall t \geq 0, \ (1+i)^t = (1+i_m)^{mt}$$

that is called the *formula for equivalent rates in compound interest* (we observe that i and i_m are linked by a non-linear relationship). We obtain the following formulas

$$\begin{cases} i = (1+i_m)^m - 1 \\ \text{Starting from } i_m \end{cases} \qquad \text{and} \qquad \begin{cases} i_m = (1+i)^{1/m} - 1 \\ \text{Starting from } i \end{cases}$$

Example 27. The semiannual interest rate i_2 equivalent on compound accumulation to the annual interest rate $i = 12\%$ is found using the above formula as follows

$$(1+i_2)^2 = 1.12 \quad \Rightarrow \quad 1 + i_2 = \sqrt{1.12} \quad \Rightarrow \quad i_2 = \sqrt{1.12} - 1 \approx 0.0583$$

The annual interest rate i equivalent on compound accumulation to the monthly interest rate $i_{12} = 0.0098$ is found using the above formula as follows

$$1 + i = (1 + 0.0098)^{12} \quad \Rightarrow \quad i = 1.0098^{12} - 1 \approx 0.1242 = 12.42\%$$

Remark 28. From now on we will indicate with j_m the *nominal annual interest rate*, convertible m times in a year (in Italy this parameter is called TAN, which means "Tasso Annuo Nominale"). This parameter must be indicated in any financial contract, and it is linked to the parameter i_m by the formula $j_m = i_m \cdot m$ (remember that we will use i_m or i for any financial calculation and not j_m). The law states that in all financial contracts (e.g. loans) the TAN must be indicated for them to be valid. We can establish a direct link between the annual interest i and the annual nominal rate j_m, in fact we have

$$\forall t \geq 0, \ \begin{cases} (1+i)^t = (1+i_m)^{mt} \\ j_m = i_m \cdot m \end{cases} \Rightarrow (1+i)^t = \left(1 + \frac{j_m}{m}\right)^{mt} \quad \forall t \geq 0$$

Example 29. A financial contract foresees the annual nominal interest rate $j_{12} = 0.1041$. Therefore the monthly interest rate i_{12} and the annual interest rate i (compound accumulation) are

$$i_{12} = \frac{0.1041}{12} = 0.008675 \qquad \text{and} \qquad i = (1 + 0.008675)^{12} - 1 = 0.1092 = 10.92\%$$

from which it emerges that $j_m \leq i$.

■ Continuously compounded interest

As we have seen above, using compound accumulation the interest capitalization dates can be annual, semi-annual, quarterly and so on. We can also consider the case when the interest capitalization is continuous (that is: the number of interest capitalization dates tends to $+\infty$). If we indicate with $\delta \geq 0$ a new financial parameter, known as *force of interest*, or continuous interest rate, we will say from now on that the formula

$$M = C \cdot e^{\delta t} \qquad \text{for any } t \geq 0 \tag{1.5}$$

gives the final value M at maturity t of a principal C invested at 0 (today), using compound accumulation with force of interest δ (or: using *continuously compounded accumulation*).

Remark 30. We can directly find the continuous interest accumulation factor as follows. Let f be a differentiable accumulation factor over $(0, +\infty)$. Suppose that the variation, indicated with Δf, when we move from s to $s + h$ (with $h \neq 0$) is proportional to $f(s)$ and h as follows:

$$\Delta f = f(s+h) - f(s) = \delta(s) f(s) h + o(h) \quad \text{as } h \longrightarrow 0,$$

where $\delta(s) \geq 0$ is a proportionality constant which depends on s, generally different from date to date. We have

$$\frac{f(s+h) - f(s)}{h} = \delta(s) f(s) \ + o(1)$$

and, using the property of differentiability of f, we get $f'(s) = \delta(s) f(s)$ that is $\dfrac{f'(s)}{f(s)} = \delta(s)$. Integrating from 0 to t and using the condition $f(0) = 1$, we have

$$\log f(s)|_0^t = \int_0^t \delta(s) \mathrm{d}s \quad \Rightarrow \quad \log f(t) - \log f(0) = \int_0^t \delta(s) \mathrm{d}s \quad \Rightarrow \quad f(t) = e^{\int_0^t \delta(s) \mathrm{d}s}$$

In the simplest case, that is if $\forall s \geq 0$, $\delta(s) = \delta$ is constant (i.e. it is a constant function), we have $f(t) = e^{\delta t}$.

Similarly, the formula defined by

$$A = S \cdot e^{-\delta t} \qquad \text{for any } t \geq 0$$

is used to calculate the present value A of a nominal value S available on T in *continuously compounded discount* (or compound discount with force of interest δ).

Example 31. The final value M at maturity $t = 3$ (years) of € 3000 invested today using compound accumulation with $\delta = 0.085$ is

$$M = 3000 \cdot f(3) = 3000 \cdot e^{0.085 \cdot 3} \approx 3871.38$$

Meanwhile the present value A of € 5000 available 15 months from now on compound discount with $\delta = 0.05$ is

$$A = 5000 \cdot \varphi\,(15/12) = 5000 \cdot e^{-0.05\cdot(15/12)} \approx 4697.07$$

Remark 32. We invest a principal C on compound accumulation with maturity t. We compare the two final values

$$M_1 = C\,(1+i)^t \quad \text{and} \quad M_2 = Ce^{\delta t}$$

where M_1 is found using compound accumulation with an annual interest rate i and M_2 is found using compound accumulation with a force of interest δ (therefore we have different assumptions about interest capitalization dates, as previously seen). Now we ask: under what condition between i and δ are M_1 and M_2 equal at maturity? Starting from the equality $M_1 = M_2$, we have

$$C\,(1+i)^t = Ce^{\delta t} \quad \Rightarrow \quad (1+i)^t = e^{\delta t} \qquad \text{for any } t \geq 0$$

that indicates which link there has to be between the parameters i and δ, in order to get the equality required. The last equality is called the *formula for equivalent rates* that links i and δ. Knowing one of the two parameters, we can find the other as follows:

$$\begin{cases} i = e^{\delta} - 1 \\ \text{Starting from } \delta \end{cases} \qquad \text{and} \qquad \begin{cases} \delta = \ln(1+i) \\ \text{Starting from } i \end{cases}$$

3.3. Decomposability

Let A and B be two different accumulation problems, with the same principal C invested at 0, accumulation factor f and lifetime $t \geq 0$, as follows:

- Accumulation problem A: we invest C from 0 to t and its final value at t is given by

$$M = Cf(t)$$

- Accumulation problem B: we invest C from 0 to s, with $s \leq t$ and accumulated value at s given by $Cf(s)$, and then we decide (immediately) to continue from s to t. The final value at t is

$$M^* = Cf(s)f\,(t-s)$$

Accumulation problems A and B give the same final value at t if and only if $M^* = M$ (otherwise one of the financial operations A and B will be more convenient), and this happens when

$$f(t) = f\,(t-s) \cdot f\,(s)$$

for any $t, s \geq 0$. This is a special property of an accumulation factor and we write its definition.

Definition 33. (*Decomposability*). Let $f(t)$ be an accumulation factor. The accumulation factor $f(t)$ is said to be *decomposable* when

$$f(t) = f\,(t-s) \cdot f\,(s) \tag{1.6}$$

for any s, t such that $0 \leq s \leq t$.

Example 34. Consider the accumulation factor $f(t) = 1 + it$, with $i > 0$ and $t \geq 0$. The first and second members of formula 1.6 are

$$f(t) = 1 + it \qquad \text{and} \qquad f(t-s) \cdot f(s) = (1 + i(t-s))(1 + is) = 1 + it + i^2 (t-s) s$$

therefore $f(t-s) \cdot f(s) > f(t)$ and f is not a decomposable financial law. Instead if we consider the following accumulation factor $f(t) = (1+i)^t$, with $i > 0$ and $t \geq 0$, we have

$$f(t) = (1+i)^t \qquad \text{and} \qquad f(t-s) \cdot f(s) = (1+i)^{t-s} (1+i)^s = (1+i)^t$$

from which we can say that f is a decomposable financial law. The financial meaning of $f(t-s) \cdot f(s)$ is the final value on t of $f(s)$€ invested at s.

Theorem 35. Let $f(t)$ be a continuous accumulation factor, with $t \geq 0$. The accumulation factor $f(t)$ is decomposable if and only if $f(t)$ is a compound accumulation law.

Proof. ($\Rightarrow$) Assume that $f(t)$ is a decomposable accumulation factor, that is $\forall t, s \geq 0$, $f(t) = f(s) \cdot f(t-s)$. From theorem 12, we have that $f(t) = e^{\delta t}$, with $\delta \geq 0$. ($\Leftarrow$) Consider the compound accumulation factor $f(t) = e^{\delta t}$, $\delta \geq 0$. We have

$$f(t) = e^{\delta t} = e^{\delta(t-s+s)} = e^{\delta(t-s)} \cdot e^{\delta s} = f(t-s) \cdot f(s)$$

and $f(t) = e^{\delta t}$, $\delta \geq 0$, is a decomposable accumulation factor.

Example 34 and Theorem 35 explain why compound interest is much more used than simple interest; and indeed, much more than any other accumulation function. From now on, we will only focus on compound interest laws.

4. Annuities

4.1. Introduction

A finite or infinite sequence of pairs with the form

$$\{(t_s, a_s)\}_{s=1}^{n} = \{(t_0, a_0), (t_1, a_1), (t_2, a_2), \ldots, (t_n, a_n)\}$$

and

$$\{(t_s, a_s)\}_{s=1}^{+\infty} = \{(t_0, a_0), (t_1, a_1), (t_2, a_2), \ldots, (t_n, a_n), \ldots\}$$

is said to be a financial operation (where date t_s is measured in years and cash flow a_s in euro). We now consider financial operations $\{(t_s, a_s)\}_{s=1}^{n/+\infty}$ with a special property on the sequence of cash flows, which are called *annuities* and *perpetuities*.

Definition 36. (*Annuity*). The sequence of cash flows $R_1, R_2, ..., R_n$ with the same sign and maturities, respectively, $t_1, t_2, ..., t_n$ is called an *annuity*. If $R_s \geq 0$, for any s, then the cash flows are all *inflows*, meanwhile if $R_s \leq 0$, for any s ($s = 1, ..., n$), then the cash flows are all *outflows*. We can describe an annuity using the following table

Maturities	t_1	t_2	$\cdots$	t_{n-1}	$t_n = T$
Cash flows	R_1	R_2	$\cdots$	R_{n-1}	R_n

or a finite sequence of pairs as follows $\{(t_s, R_s)\}_{s=1}^{n}$ (usually $t_0 = 0$ and $t_n = T$, that is the last maturity where we have the cash flow R_n).

Example 37. The financial operations given by

Maturities	1	2	3	4
CF	100	150	0	200

Maturities	1	2	4	5
CF	100	150	200	300

are two annuities, in fact in both cases the sequence of cash flows is positive.

Now we will see two special cases of annuities: (a) ordinary annuities; (b) due annuities.

4.2. Ordinary annuities

Definition 38. (*Ordinary annuity*). Let s be a natural number. The sequence of cash flows $R_1, R_2, ..., R_n$ with the same sign and maturities, respectively, $1, 2, ..., n$ is called an *ordinary annuity*. If $R_s \geq 0$, for any s, then the cash flows are all *inflows*, meanwhile if $R_s \leq 0$, for any s, then the cash flows are all *outflows*. We can describe an ordinary annuity using the following table

Maturities	0	1	2	$\cdots$	$n-1$	n
Cash flows		R_1	R_2	$\cdots$	R_{n-1}	R_n

Example 39. Consider the following ordinary annuities

A

Years	1	2	3	4	5
CF	100	100	100	100	100

B

Years	1	2	3	4	5
CF	100	150	250	200	300

with the same maturities but with different sequences of cash flows, namely annuity A has equal cash flows while B has cash flows that are not equal.

Ordinary annuity can be classified as ordinary annuities with constant cash flows (all the cash flows are equal to the same amount), or ordinary annuities with variable cash flows (at least one cash flow is not equal to the others).

We can calculate the value of an ordinary annuity at any time t between 0 (today) and n (the last maturity of the annuity). Using an accumulation factor $f(t)$, the final value M of the annuity at maturity t_n (i.e. $t_n = T$) is

$$M = R_1 \cdot f(n-1) + R_2 \cdot f(n-2) + \cdots + R_n \cdot f(n-n) = \sum_{s=1}^{n} R_s \, f(n-s)$$

that is the sum of the final values of all the cash flows at maturity n. Instead, using a discount factor $\varphi(t)$, the present value A of the annuity is

$$A = R_1 \cdot \varphi(1) + R_2 \cdot \varphi(2) + \cdots + R_{n-1} \cdot \varphi(n-1) + R_n \cdot \varphi(n) = \sum_{s=1}^{n} R_s \, \varphi(s)$$

Example 40. An ordinary annuity is described by the following table

Maturities	1	2	3
Cash flows	100	150	200

Using the annual compound interest rate $i = 6\%$, the final value M at maturity $T = 5$ is

$$M = \sum_{s=1}^{3} R_s \, 1.06^{5-s} = 100 \cdot 1.06^2 + 150 \cdot 1.06^1 + 200 \approx 4471.36$$

and the present value A (the simple sum of the present values of each cash flow) is

$$A = \sum_{s=1}^{3} \frac{R_s}{1.06^s} = \frac{100}{1.06} + \frac{150}{1.06^2} + \frac{200}{1.06^3} \approx 395.76$$

The value of the ordinary annuity at $t = 1.5$, indicated with $V_{1.5}$, is

$$V_{1.5} = 100 \cdot 1.06^{0.5} + \frac{150}{1.06^{0.5}} + \frac{200}{1.06^{1.5}} \approx 431.91$$

Remark 41. We recall that the sum of n terms of a geometric sequence with general term $(1+i)^{-s}$ is

$$\begin{aligned} \sum_{s=1}^{n} \left(\frac{1}{1+i}\right)^s &= \sum_{s=0}^{n-1} \left(\frac{1}{1+i}\right)^{s+1} = \frac{1}{1+i} \cdot \sum_{s=0}^{n-1} \left(\frac{1}{1+i}\right)^s = \frac{1}{1+i} \cdot \frac{1-(1+i)^{-n}}{1-(1+i)^{-1}} \\ &= \frac{1-(1+i)^{-n}}{(1+i) \cdot \left[1-(1+i)^{-1}\right]} = \frac{1-(1+i)^{-n}}{(1+i)-1} = \frac{1-(1+i)^{-n}}{i} \end{aligned}$$

Remark 42. We want to calculate the value of an ordinary annuity under the following assumptions: (a) the dates are years (that is the cash flow is at the end of each year); (b) we use compound financial factors in the evaluation; (c) the cash flows are all equal, that is $R_s = R$ with $s = 1, 2, ..., n$.
The final value M at maturity $T = n$ is

$$M = \sum_{s=1}^{n} R\,(1+i)^{n-s} = R\,(1+i)^n \sum_{s=1}^{n} \left(\frac{1}{1+i}\right)^s = R \cdot \frac{(1+i)^n - 1}{i} = R \cdot s_{\overline{n}|i}$$

where from now on we will indicate the ratio $[(1+i)^n - 1]/i$ with the financial symbol $s_{\overline{n}|i}$ (which is read as "s ordinary n angle i" or "s ordinary n corner i", where n is the number of cash flows and i is the annual interest rate). Meanwhile, the present value A is

$$A = \sum_{s=1}^{n} R\,(1+i)^{-s} = R \sum_{s=1}^{n} \left(\frac{1}{1+i}\right)^s = R \cdot \frac{1-(1+i)^{-n}}{i} = R \cdot a_{\overline{n}|i}$$

where from now on we will indicate the ratio $\left[1-(1+i)^{-n}\right]/i$ with the financial symbol $a_{\overline{n}|i}$ (which is read as "a ordinary n angle i" or "a ordinary n corner i").

Example 43. An ordinary annuity is described by the following table

Years	1	2	3	4	5	6
Cash flows	1500	1500	1500	1500	1500	1500

Using compound annual interest rate $i = 8\%$, the final value M at maturity $T = 6$ is

$$M = 1500 \cdot s_{\overline{6}|0.08} = 1500 \cdot \frac{1.08^6 - 1}{0.08} \approx 11003.89$$

and the present value A is

$$A = 1500 \cdot a_{\overline{6}|0.08} = 1500 \cdot \frac{1 - 1.08^{-6}}{0.08} \approx 6934.32$$

4.3. Due annuities

Definition 44. (*Due annuity*). Let s be a natural number. The sequence of cash flows $R_1, R_2, ..., R_n$ with the same sign and maturities, respectively, $0, 1, ..., n-1$ is called a *due annuity*. If $R_s \geq 0$, for any s, then the cash flows are all *inflows*, meanwhile if $R_s \leq 0$, for any s, then the cash flows are all *outflows*. We can describe a due annuity using the following table

Maturities	0	1	2	$\cdots$	$n-1$	n
Cash flows	R_1	R_2	R_3	$\cdots$	R_n	

or a finite sequence of pairs as follows $\{(t_s, R_s)\}_{s=0}^{n}$.

We can calculate the value of a due annuity at any time t between 0 (today) and $t_n = n$ (the last maturity of the annuity). Using an accumulation factor $f(t)$, the final value M of the annuity at maturity t_n is

$$M = R_1 \cdot f(n-0) + R_2 \cdot f(n-1) + \cdots + R_n \cdot f(n-(n-1)) = \sum_{s=1}^{n} R_s\, f(n-(s-1))$$

that is the sum of the final values of all the cash flows at maturity n. Instead, using a discount factor $\varphi(t)$, the present value A of the due annuity is

$$A = R_1 \cdot \varphi(0) + R_2 \cdot \varphi(1) + \cdots + R_{n-1} \cdot \varphi(n-2) + R_n \cdot \varphi(n-1) = \sum_{s=1}^{n} R_s\, \varphi(s-1)$$

Example 45. A due annuity, with maturity 5 years, is described by the following table

Maturities	0	1.5	4
Cash flows	1000	2500	2000

Using the annual compound interest rate $i = 6\%$, final value M at maturity $T = 5$ is

$$M = \sum_{s=1}^{3} R_s (1+0.06)^{5-t_{s-1}} = 1000 \cdot 1.06^5 + 2500 \cdot 1.06^{3.5} + 2000 \cdot 1.06 \approx 6523.79$$

and the present value A (the sum of the present values of each cash flow) is

$$A = \sum_{s=1}^{3} \frac{R_s}{(1+0.06)^{t_{s-1}}} = 1000 + \frac{2500}{1.06^{1.5}} + \frac{2000}{1.06^4} \approx 4874.96$$

The value on $t = 2$ of the due annuity, indicated with V_2, is

$$V_2 = 1000 \cdot 1.06^2 + 1500 \cdot 1.06^{0.5} + \frac{2000}{1.06^2} \approx 4447.94$$

Remark 46. We want to calculate the value of a due annuity under the following assumptions: (1) the maturities are years (i.e. the cash flow is at the end of each year); (2) we use compound financial factors in the evaluation; (3) the cash flows are all equal, that is $R_s = R$ with $s = 1, 2, ..., n$. The final value M at maturity $T = n$ is

$$M = \sum_{s=1}^{n} R(1+i)^{n-(s-1)} = R(1+i)^n \sum_{s=1}^{n} \left(\frac{1}{1+i}\right)^{s-1} = R \cdot (1+i) \cdot \frac{(1+i)^n - 1}{i} = R \cdot \ddot{s}_{\overline{n}|i}$$

where from now on we will indicate product $(1+i)\cdot\left[(1+i)^n-1\right]/i$ with the financial symbol $\ddot{s}_{\overline{n}|i}$ (which is read as "s due n angle i", where n is the number of the cash flows and i is the annual interest rate). Meanwhile, the present value A is

$$A=\sum_{s=1}^{n} R\,(1+i)^{-(s-1)}=R\,(1+i)\sum_{s=1}^{n}\left(\frac{1}{1+i}\right)^{s}=R\cdot(1+i)\cdot\frac{1-(1+i)^{-n}}{i}=R\cdot\ddot{a}_{\overline{n}|i}$$

where from now on we will indicate product $(1+i)\cdot\left[1-(1+i)^{-n}\right]/i$ with the financial symbol $\ddot{a}_{\overline{n}|i}$ (which is read as "a due n angle i").

4.4. Ordinary and due perpetuities

Definition 47. (*Ordinary perpetuity*). A financial operation described by the following table

Years	$t_0=0$	1	2	3	$\cdots$	$n-1$	n	$\cdots$
Cash flows		R	R	R	$\cdots$	R	R	$\cdots$

is called an *ordinary perpetuity* (where the cash flows are infinitely many, and for simplicity we suppose that all of them have the same value).

Remark 48. In compound discount, the present value A of an ordinary perpetuity is defined by

$$A=\frac{R}{1+i}+\frac{R}{(1+i)^2}+\frac{R}{(1+i)^3}+\cdots+\frac{R}{(1+i)^{n-1}}+\frac{R}{(1+i)^n}+\cdots$$

therefore

$$\begin{aligned}A&=\sum_{s=1}^{+\infty} R\,(1+i)^{-s}=R\lim_{n\to+\infty}\sum_{s=1}^{n}(1+i)^{-s}=R\lim_{n\to+\infty}\sum_{s=1}^{n}\left(\frac{1}{1+i}\right)^{s}\\&=R\cdot\lim_{n\to+\infty}\frac{1-(1+i)^{-n}}{i}=\frac{R}{i}=R\cdot a_{\overline{\infty}|i}\quad\textbf{N.B. }(1+i)^{-n}\to0\text{ as }n\to+\infty\end{aligned}$$

where from now on we will indicate the ratio $1/i$ with the financial symbol $a_{\overline{\infty}|i}$ (which is read as "a ordinary ∞ angle i").

Example 49. Using an annual compound interest rate $i=4\%$, we want to calculate the present value A of the following ordinary perpetuity

Years	1	2	3	4	5	6	7	8	9	10
Payments	500	500	500	500	500	1000	$\cdots$	$\cdots$	1000	$\cdots$

We can consider the above perpetuity as a union of an ordinary annuity (with five constant payments of € 500) and a perpetuity (with its first cash flow 6 years later). The present value A is

$$A=500\cdot a_{\overline{5}|0.04}+\frac{1000}{0.04}\cdot1.04^{-5}=500\cdot\frac{1-1.04^{-5}}{0.04}+\frac{1000}{0.04}\cdot1.04^{-5}\approx22774.09$$

Definition 50. (*Due perpetuity*). A financial operation described by the following table

Years	$t_0=0$	1	2	3	$\cdots$	$n-1$	n	$\cdots$
Cash flows	R	R	R	R	$\cdots$	R	R	$\cdots$

is called a *due perpetuity* (where the cash flows are infinitely many, and for simplicity we suppose that all of them have the same value).

Remark 51. In compound discount, the present value A of an ordinary perpetuity is defined by

$$A = R + \frac{R}{1+i} + \frac{R}{(1+i)^2} + \frac{R}{(1+i)^3} + \cdots + \frac{R}{(1+i)^{n-1}} + \frac{R}{(1+i)^n} + \cdots$$

therefore

$$\begin{aligned} A &= \sum_{s=0}^{+\infty} R\,(1+i)^{-s} = R \lim_{n\to+\infty} \sum_{s=0}^{n} (1+i)^{-s} = R \cdot \lim_{n\to+\infty} \left[1 + \sum_{s=1}^{n} (1+i)^{-s}\right] \\ &= R + \frac{R}{i} = R \cdot \frac{1+i}{i} = R \cdot \ddot{a}_{\overline{\infty}|i} \end{aligned}$$

where, from now on, we will indicate the ratio $(1+i)/i$ with the financial symbol $\ddot{a}_{\overline{\infty}|i}$ (which is read as "a due ∞ angle i").

5. Discounted Cash Flow

5.1. Discounted Cash Flow and Net Present Value

Consider the following financial operation

Years	$t_0 = 0$	t_1	t_2	$\cdots$	t_{n-1}	$t_n = T$
Cash flows	a_0	a_1	a_2	$\cdots$	a_{n-1}	a_n

The function $G : (-1, +\infty) \to \mathbb{R}$ defined by

$$G(x) = \sum_{s=0}^{n} \frac{a_s}{(1+x)^{t_s}} = \frac{a_0}{(1+x)^{t_0}} + \frac{a_1}{(1+x)^{t_1}} + \frac{a_2}{(1+x)^{t_2}} + \cdots + \frac{a_n}{(1+x)^{t_n}}$$

is called the *Discounted Cash Flow (DCF)* associated to the financial operation with an annual interest rate $x > -1$ (please note: we now consider Axiom A.3°). If we fix an annual interest rate x^* then we can calculate the *DCF* at x^* as follows

$$G(x^*) = \sum_{s=0}^{n} \frac{a_s}{(1+x^*)^{t_s}} = \frac{a_0}{(1+x^*)^{t_0}} + \frac{a_1}{(1+x^*)^{t_1}} + \frac{a_2}{(1+x^*)^{t_2}} + \cdots + \frac{a_n}{(1+x^*)^{t_n}}$$

which is called the *Net Present Value (NPV)* at x^* of the financial operation. The difference between DCF and NPV of a financial operation is clear: *DCF* is a function of $x > -1$ and NPV is a real number (positive, negative or null).
From now on a financial operation, with maturity T, described by the following table

Years	$t_0 = 0$	t_1	t_2	t_3	$\cdots$	t_{n-1}	$t_n = T$
Cash flows	a_0	a_1	a_2	a_3	$\cdots$	a_{n-1}	a_n

is called:

(a) an *investment* if $a_0 < 0$ and $a_s \geq 0$, $\forall s = 1, 2, ..., n$, with at least an $a_s > 0$;

(b) a *loan* if $a_0 > 0$ and $a_s \leq 0$, $\forall s = 1, 2, ..., n,$, with at least an $a_s < 0$.

Example 52. An investment with a 2-year maturity is described in the following table

Years	0	1	3
Cash flows	-1500	300	2000

The Discounted Cash Flow of the investment, with an annual interest rate $x > -1$, is

$$G(x) = \sum_{s=0}^{2} \frac{a_s}{(1+x)^{t_s}} = -1500 + \frac{300}{1+x} + \frac{2000}{(1+x)^3}$$

The net present values at $x = 3\%$ and $x = 21\%$ are

$$G(0.03) = \sum_{s=0}^{2} \frac{a_s}{1.03^{t_s}} \approx 621.55 > 0 \qquad \text{and} \qquad G(0.21) = \sum_{s=0}^{2} \frac{a_s}{1.21^{t_s}} \approx -123.1 < 0$$

In special cases we can immediately draw the graph of the DCF of the financial operation.

- If a financial operation has $a_0 < 0$ and $a_s \geq 0$ (with at least one flow a_s being strictly positive), then the associated DCF has the following properties:

  (a) $G(x) \to +\infty$ as $x \to -1^+$ ($x = -1$ is a vertical asymptote) and $G(x) \to a_0$ as $x \to +\infty$ ($y = a_0 < 0$ is a horizontal asymptote);

  (b) Function G is differentiable on $(-1, +\infty)$ and

  $$G'(x) = \frac{\mathrm{d}}{\mathrm{d}x} \sum_{s=0}^{n} \frac{a_s}{(1+x)^{t_s}} = -\sum_{s=1}^{n} a_s t_s (1+x)^{-t_s} < 0$$

  for any $x > -1$, therefore $G(x)$ is strictly decreasing on $(-1, +\infty)$;

  (c) G has a unique point of intersection x^* with the x-axis such that $G(x^*) = 0$;

  (d) $G''(x) > 0$ for any $x > -1$, therefore $G(x)$ is strictly convex on $(-1, +\infty)$.

  Figure 5 shows the graph of a DCF of a financial operation with the features indicated.

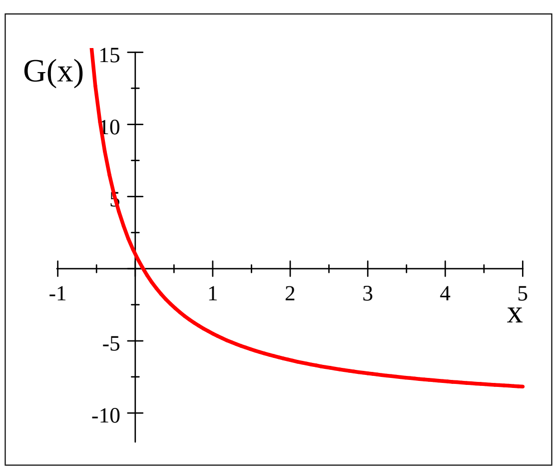

Figure 5

- If a financial operation has $a_0 > 0$ and $a_s \leq 0$ (with at least one flow a_s being strictly negative) then the associated DCF has the following properties:

  (a) $G(x) \to -\infty$ as $x \to -1^+$ ($x = -1$ is a vertical asymptote) and $G(x) \to a_0$ as $x \to +\infty$ ($y = a_0 > 0$ is a horizontal asymptote);

  (b) Function G is differentiable on $(-1, +\infty)$ and

$$G'(x) = \frac{\mathrm{d}}{\mathrm{d}x} \sum_{s=0}^{n} \frac{a_s}{(1+x)^{t_s}} = -\sum_{s=1}^{n} a_s t_s (1+x)^{-t_s} > 0$$

  for any $x > -1$, therefore $G(x)$ is strictly increasing on $(-1, +\infty)$;

  (c) G has a unique point of intersection x^* with the x-axis such that $G(x^*) = 0$;

  (d) $G''(x) < 0$ for any $x > -1$, therefore $G(x)$ is strictly concave on $(-1, +\infty)$.

  Figure 6 shows the graph of a DCF of a financial operation with the features indicated.

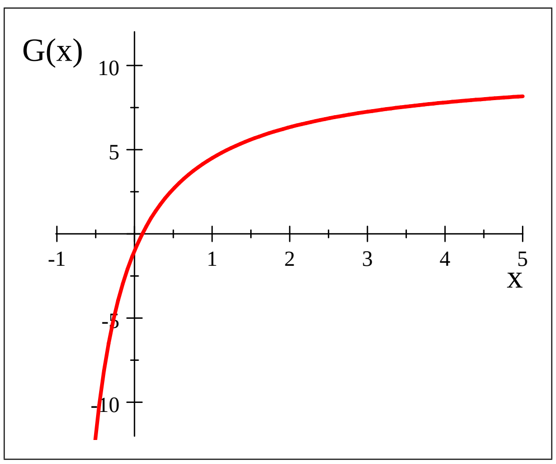

Figure 6

5.2. Internal Rates of a financial operation

We are often interested in finding for a financial operation which annual interest rates, if any, produce a null DCF. These annual interest rates (if they exist, be careful!) are called the *internal rates* of the financial operation. To find them we must solve the equation $G(x) = 0$. In special situations (investments and loans) we are certain that the internal rate exists and is unique (sometimes this unique annual interest rate is called the *implicit rate* of the financial operation).

Definition 53. (*internal rate of return, effective cost*). The internal rate of an investment is called the *internal rate of return* (of the investment), or *IRR*, while the internal rate of a loan is called the *effective cost* of the loan.

Example 54. Consider investment A with a 2-year maturity described by the following table

Years	0	1	2
Cash flows	-1000	700	600

where its discounted cash flow, with annual interest rate $x > -1$, is

$$G(x) = \sum_{s=0}^{2} \frac{a_s}{(1+x)^s} = -1000 + \frac{700}{1+x} + \frac{600}{(1+x)^2}$$

As it is an investment we have all the properties mentioned above ($a_0 < 0$ and $a_1, a_2 > 0$). The NPV of the investment at $x_1 = 8\%$ is

$$G(0.08) = -1000 + \frac{700}{1+0.08} + \frac{600}{(1+0.08)^2} \approx 162.55 > 0$$

Indicating with x^* the unique IRR of investment A, we observe that

$$\begin{cases} \text{Investment } A \Rightarrow G \text{ is strictly decreasing} \\ G(0.08) \approx 162.55 > 0 \end{cases} \quad \Rightarrow \quad \text{the IRR } x^* > x_1 = 0.08$$

Although we do not know its precise value we can say that it will be greater than 8%. The NPV of the investment at $x_2 = 23\%$ is

$$G(0.23) = -1000 + \frac{700}{1+0.23} + \frac{600}{(1+0.23)^2} \approx -34.30 < 0$$

therefore

$$\begin{cases} \text{Investment } A \Rightarrow G \text{ is strictly decreasing} \\ G(0.23) \approx -34.30 < 0 \end{cases} \quad \Rightarrow \quad \text{the IRR } x^* < x_2 = 0.23$$

Although we do not know its precise value we can say that it will be lower than 23%.

6. Fixed-income bonds

A *fixed-income bond* is a security that contractually guarantees future payments both in terms of the amounts and of the maturity dates. Stocks do not fall into this category: only bonds can be considered fixed-income.
Purchasing a fixed-income bond, therefore, should be a way to invest without risk. Actually, there always exists a risk of insolvency (*default risk*) linked to the entity that issued the bonds: for this reason even fixed income bonds are classified on the financial markets using a *rating* that depends on the credibility of the issuing entity; however, we will not enter into details on this issue.
We have two types of fixed-income bonds:

(i) *zero coupon bond (zcb)*

(ii) *bonds with coupons*

For both of them we will define the Yield To Maturity, which is nothing but the internal rate of return of the investment given by purchasing the bond.
A *zero coupon bond (zcb)* is the simplest case of fixed-income bonds. It is characterized by

Years	0	T
Cash flows	$-P_0$	N

where

- $P_0 =$ price of the zcb at $t_0 = 0$;
- $T =$ maturity of the zcb;
- $N =$ face (or nominal) value.

We have

Definition 55. (*Yield To Maturity of a zcb*). The *Yield to Maturity (YTM)*, or gross compound yield, of a zcb is the gross compound rate of return $i = i(0,T)$ that characterizes the investment from 0 until the maturity $T \geq 0$ of the zcb

$$i(0,T) = \left(\frac{N}{P_0}\right)^{\frac{1}{T}} - 1$$

Note that knowing the YTM i of a zcb allows to calculate its price today

$$P_0 = N(1+i)^{-T}$$

Example 56. Consider a zcb with face value $N = 1010$, maturity $T = 3$ months and today price $P_0 = 1000$. Its YTM is

$$i = \left(\frac{1010}{1000}\right)^{1/4} - 1 \approx 0.025$$

Some remarks:

- The YTM is a "gross" return rate, since it does not consider any taxation. In case taxes are considered, we get *net* return rates.
- The YTM of a zcb may change during the zcb lifetime. Thereby we may compute $i(t,T)$ for every $t \in [0,T]$ by knowing P_t.
- The yield on zcbs can also be measured using annual simple interest rates. For example, this is the case of Italian BOTs.
- When measuring returns on bonds, maturities are expressed in days. So, the maturities typically have the format $T = dd/365$ where dd is the number of days remaining until the redemption.

In the category of fixed-income bonds we also have *bonds with coupons* (we suppose that every coupon is fixed once and for all, at time 0; there could also be more general situations, but we do not enter into details). These are bonds which, in addition to a reimbursement value R at a maturity T, pay periodic amounts called *coupons*. We will consider bonds paying coupons $c_1, ..., c_n > 0$ at the maturities $t_1, ..., t_n = T$:

Years	$t_0 = 0$	t_1	t_2	$\cdots$	$t_n = T$
Cash flows	$-P_0$	c_1	c_2	$\cdots$	$c_n + R$

In particular, it can be $c_1 = ... = c_n = c$: we obtain bonds with constant coupons.
There are various quantities that are used to describe a bond with coupons.

- $t_k =$ periodic coupon maturities, $k = 1, 2, \cdots, n$. There are two typical cases:

$$\pi = t_k - t_{k-1} = \begin{cases} 1/2 \Longrightarrow \text{ semi-annual coupons} \\ 1 \Longrightarrow \text{ annual coupons} \end{cases} \qquad \forall k = 1, 2, \cdots, n$$

- $T = t_n =$ final maturity of the bond;
- $R =$ reimbursement value;
- $c_1, ..., c_n =$ coupons;
- $P_0 =$ the market price of the bond at date $t_0 = 0$.

Also in the case of coupon bonds, the yield is measured by its compound rate of return.

Definition 57. (*Yield To Maturity of a bond with coupons*). The *Yield To Maturity (YTM)* of the bond described by the cash flow

Years	$t_0 = 0$	t_1	t_2	$\cdots$	$t_n = T$
Cash flows	$-P_0$	c_1	c_2	$\cdots$	$c_n + R$

is the compound interest rate x^* which solves the equation

$$G(x) = -P_0 + \sum_{s=1}^{n} c_s (1+x)^{-t_s} + R(1+x)^{-t_n} = 0$$

Let us consider an example.

Remark 58. If the bond cash flow is

Years	0	π	2π	$\cdots$	$n\pi$
Cash flows	$-P$	c	c	$\cdots$	$c+P$

then its YTM is

$$x^* = \begin{cases} c/P & \text{if } \pi = 1 \\ (1+c/P)^2 - 1 & \text{if } \pi = 1/2 \end{cases}$$

7. Duration

We begin by presenting the following

Definition 59. (*Duration*). Given the cash flow

(A)

Years	t_1	t_2	$\cdots$	t_n
Inflows	a_1	a_2	$\cdots$	a_n

with $a_s > 0, s = 1, \cdots, n$, the *(Macaulay) Duration* of (A), calculated at the (annual compound interest) rate $i > -1$ is the quantity

$$\boxed{D(i) = \frac{\sum_{s=1}^{n} t_s \cdot a_s (1+i)^{-t_s}}{\sum_{s=1}^{n} a_s (1+i)^{-t_s}}}$$

Observe that:

- The Duration is the arithmetic mean of the maturites t_s weighted with the discounted values $p_s = a_s (1+i)^{-t_s}$ of each inflow a_s at each maturity, calculated using the annual compound interest rate i:

  $$D(i) = \frac{\sum_{s=1}^{n} t_s \cdot p_s}{\sum_{s=1}^{n} p_s}$$

  From $a_s > 0$ it follows that $p_s > 0$ for every $i > -1$: the weights used in calculating D are always strictly positive.

- Duration is always expressed using the same unit of time as t, which is in years by default.
- We always have that
$$t_1 \leq D(i) \leq t_n \ \text{ for every } i > -1$$
In particular, if there is only one flow at maturity T then $D(i) = T$ for every $i > -1$.
- It can be shown that in general, $D(i)$ is a decreasing function of i and that
$$D(i) \to t_n \text{ as } i \to -1^+ \ \text{ and } \ D(i) \to t_1 \text{ as } i \to +\infty$$

Duration is typically used when referring to the revenue flow derived from an investment. We will focus on the specific case of *fixed-income bonds.*
Therefore, the flow (A) represents the revenues derived from the purchase of a bond (or a portfolio of bonds) at $t_0 = 0$ with price P_0 and $D(i)$ is the Duration of bond (A).
We observe that if i is the *yield to maturity* of the bond, then:

- The price of the bond calculated at rate i, which is the price on the market today, is $P_0 = \sum_{s=1}^{n} a_s (1+i)^{-t_s} = P(i)$, which is exactly the denominator of the Duration:
$$D(i) = \frac{\sum_{s=1}^{n} t_s \cdot a_s (1+i)^{-t_s}}{P(i)}$$
- The rate i coincides with the IRR x^* of the investment, since
$$G(i) = -P_0 + \sum_{s=1}^{n} a_s (1+i)^{-t_s} = 0$$

Example 60. The coupon bond described by:

Years	0	1	2	3
Cash flows	-1000	50	50	1050

has yield to maturity $x^* = 0.05$. The duration of the bond at annual compound interest rate $i = 5\% = x^*$ is
$$D(0.05) = \frac{1 \cdot 50 \cdot 1.05^{-1} + 2 \cdot 50 \cdot 1.05^{-2} + 3 \cdot 1050 \cdot 1.05^{-3}}{1000} \approx 2.8594 \text{ years}$$

7.1. Financial immunization

Suppose that we would like to invest an amount of money at $t = 0$ until the time $z > 0$ at today's yield rate i. To invest, we can simply purchase a coupon bond with YTM i and maturity $t_n \geq z$. To analyze the investment we consider here two different scenarios.

Scenario I − The rate i stays constant for the lifetime of the investment: i is the annual compound interest rate valid for the entire time horizon $[0, z]$. In this case:

(i) The principal invested today at $t = 0$ coincides with the market price paid for the purchase of the bond
$$P_0 = \sum_{s=1}^{n} a_s (1+i)^{-t_s}$$

(ii) As each of the coupons matures, we reinvest them at the rate i until time z. From the reinvestment of the coupons at rate i we obtain the future value at time z

$$M(z,i) = \sum_{t_k<z} a_k (1+i)^{z-t_k}$$

(iii) In z years, we realize the investment by selling the bond on the market (in fact, this sale can occur at any time $z \leq t_n$) and collect the price that coincides with the value of the remaining flows discounted to time z at the same rate i:

$$P(z,i) = \sum_{t_k \geq z} a_k (1+i)^{-(t_k-z)} = \sum_{t_k \geq z} a_k (1+i)^{z-t_k}$$

(iv) The final result of the investment at z is therefore the sum of the two reinvestment/price parts:

$$V(z,i) = M(z,i) + P(z,i) = \sum_{t_k \leq z} a_k (1+i)^{z-t_k} + \sum_{t_k > z} a_k (1+i)^{z-t_k} =$$

$$= \sum_{k=1}^{n} a_k (1+i)^{z-t_k} = (1+i)^z \sum_{k=1}^{n} a_k (1+i)^{-t_k} = P_0 (1+i)^z$$

In fact, the final result of the investment

$$(*) \qquad V(z,i) = \sum_{k=1}^{n} a_k (1+i)^{z-t_k} = (1+i)^z \sum_{k=1}^{n} a_k (1+i)^{-t_k} = P_0 (1+i)^z$$

coincides with the future value of the invested amount P_0 after z years at a rate of i.

Scenario II – The rate i undergoes a variation Δi at $\tau \in (0, t_1]$, and the new rate $i' = i + \Delta i$ remains unchanged from τ onward (Please note: we do not cover here the case of more than one variation). What will happen in this scenario?
Given that the market rate variation occurred after the bond purchase, the principal remains as P_0. What changes is the final result of the investment which must be calculated using the new rate $i' = i + \Delta i$:

$$V(z, i+\Delta i) = M(z, i+\Delta i) + P(z, i+\Delta i)$$

Therefore, we obtain the following:
(1) The relation (*) no longer holds, i.e.

$$V(z, i+\Delta i) \neq P_0 (1+i)^z = V(z,i)$$

(2) The difference $\Delta V(z,i) = V(z, i+\Delta i) - V(z,i)$ between the result obtained and the result expected is the sum of the variations of the two addends

$$\Delta V(z,i) = \Delta M(z,i) + \Delta P(z,i)$$

which partially offset each other because they always have opposite signs.

$$\Delta i > 0 \Longrightarrow \Delta M\left(z, i\right) > 0 \ \text{ and } \ \Delta P\left(z, i\right) < 0$$
$$\Delta i < 0 \Longrightarrow \Delta M\left(z, i\right) < 0 \ \text{ and } \ \Delta P\left(z, i\right) > 0$$

(3) The sign of the overall variation $\Delta V\left(z, i\right)$ is thus unpredictable since it depends on the nature of both underlying variations.
The possibility of having negative variations is called *interest rate risk*; it is composed of two risks of opposite sign, called *reinvestment risk* and *price risk.*
We observe that, in addition to Δi, the extent of the variation of the two components depends on the maturity z of the investment:

- If z is sufficiently close to the maturity t_n of the bond, there is little to discount and a lot to accumulate $\Longrightarrow$ the reinvestment component is more influential: $|\Delta M| > |\Delta P|$.
- If z is sufficiently close to the maturity of the first coupon at t_1, there is a lot to discount and little to accumulate $\Longrightarrow$ the price component is more influential: $|\Delta P| > |\Delta M|$.

Following this reasoning, there should exist an intermediate maturity z^* at which the positive component is more substantial, thus eliminating the interest rate risk.
In fact, this maturity does exist. First of all, we will clarify what we mean by the following

Definition 61. (*Financial immunization*). An investment at rate i is *financially* (globally) *immunized at year* z^* against the interest rate risk if

$$\boxed{V\left(z^*, i + \Delta i\right) \geq V\left(z^*, i\right) \quad \text{for every } \Delta i}$$

and the lifetime z^* is called the *date of immunization.*

Stated differently, the date of immunization z^* is the lifetime that makes the initial rate i the point of *global minimum* of the function $f\left(x\right) = V\left(z^*, x\right)$ as the annual compound interest rate $x = i + \Delta i$ varies. It is for this reason that we speak of *global* immunization.
Under the hypotheses we have already seen, it can be proven that the duration of an investment is, in fact, the date of immunization.

Theorem 62. Given a fixed income bond that pays

Years	t_1	t_2	$\cdots$	t_n
Inflows	a_1	a_2	$\cdots$	a_n

if (i) the yield to maturity of the bond at $t_0 = 0$ is i; (ii) the rate i undergoes a variation Δi at $\tau \in (0, t_1]$ and the new rate $i' = i + \Delta i$ does not undergo further variations;
then the bond investment is financially immunized against the interest rate risk at the date

$$\boxed{z^* = \frac{\sum_{s=1}^{n} t_s \cdot a_s \left(1 + i\right)^{-t_s}}{\sum_{s=1}^{n} a_s \left(1 + i\right)^{-t_s}} = D\left(i\right)}$$

Proof. We want to find the time z^* such that the minimizer with respect to the rate x of the function

$$f\left(x\right) = V\left(z^*, x\right) = \sum_{k=1}^{n} a_k \left(1 + x\right)^{z^* - t_k}$$

is $x^* = i$. The function f is differentiable on $(-1, +\infty)$ with derivative $f' = \partial V/\partial x$

$$f'(x) = \sum_{k=1}^{n} a_k \cdot (z^* - t_k)(1+x)^{z^*-t_k-1} = (1+x)^{z^*-1} \sum_{k=1}^{n} a_k \cdot (z^* - t_k)(1+x)^{-t_k} =$$
$$= (1+x)^{z^*-1} \left[z^* \sum_{k=1}^{n} a_k \cdot (1+x)^{-t_k} - \sum_{k=1}^{n} a_k \cdot t_k (1+x)^{-t_k} \right]$$

Since every minimizer x^* of f is a stationary point, if we want the minimizer to be $x^* = i$ we must have

$$f'(i) = 0 \Longleftrightarrow z^* = \frac{\sum_{k=1}^{n} t_k \cdot a_k (1+i)^{-t_k}}{\sum_{k=1}^{n} a_k \cdot (1+i)^{-t_k}} = D(i)$$

Let us now consider the second derivative

$$\begin{aligned} f''(x) &= \sum_{k=1}^{n} a_k \cdot (z^* - t_k)(z^* - t_k - 1)(1+x)^{z^*-t_k-2} \\ &= \sum_{k=1}^{n} a_k \cdot (z^* - t_k)^2 (1+x)^{z^*-t_k-2} - \sum_{k=1}^{n} a_k \cdot (z^* - t_k)(1+x)^{z^*-t_k-2} \\ &= \sum_{k=1}^{n} a_k \cdot (z^* - t_k)^2 (1+x)^{z^*-t_k-2} - (1+x)^{-1} f'(x) \end{aligned}$$

If we calculate $f''(x)$ at the stationary point $x^* = i$ we get

$$f''(i) = \sum_{k=1}^{n} a_k \cdot (z^* - t_k)^2 (1+x)^{z^*-t_k-2}$$

Since each term is positive and at least one of them is strictly positive if $n > 1$, it is $f''(i) > 0$. This proves that i is an (at least) local minimizer for f.

Some observations:

- The proof we just provided is not complete, since we did not explicitly prove that the stationary point $x^* = i$ is a *global* minimizer. This can be completed in many ways, for example by showing that the function $f(x) = V(z^*, x)$ is convex, and consequently any stationary point is a global minimizer. We often use the term *convexity* of the bond, referring to the convexity of $f(x)$, which is usually established by showing that $f''(x) = \partial^2 V/\partial x^2 > 0$.

- Strictly speaking, any variation Δi that occurs after today is random. Despite this, we have treated Δi as a real variable: this is a *semi-deterministic approach* to a problem that is essentially random in nature.

- Note that for a zcb whose maturity is T, the duration, and therefore the date of immunization, coincides with the remaining lifetime of the bond: $D(i) = T$ for every i. To be immunized against interest rate risk, we must hold the zcb until maturity.

If an investment is immunized, then any variation Δi is welcome: it will lead to a higher return than expected.

Example 63. We consider again the previous investment at 5%:

Years	0	1	2	3
Cash flows	-1000	50	50	1050

with the date of immunization

$$D(0.05) = \frac{1 \cdot 50 \cdot 1.05^{-1} + 2 \cdot 50 \cdot 1.05^{-2} + 3 \cdot 1050 \cdot 1.05^{-3}}{1000} \approx 2.8594 \text{ years}$$

If we redeem the investment at $z^* = 2.8594$ years we obtain the expected amount

$$\begin{aligned}
V(z^*, 0.05) &= 50 \cdot 1.05^{2.8594-1} + 50 \cdot 1.05^{2.8594-2} + 1050 \cdot 1.05^{2.8594-3} \\
&= 1.05^{2.8594} \left(50 \cdot 1.05^{-1} + 50 \cdot 1.05^{-2} + 1050 \cdot 1.05^{-3}\right) \\
&= 1.05^{2.8594} \cdot 1000 \\
&\approx \boxed{1149.7110}
\end{aligned}$$

If within the first year YTM undergoes a variation of $\Delta i = 2\%$, the result obtained at $z^* = 2.8594$ years becomes

$$\begin{aligned}
V(z^*, 0.07) &= 50 \cdot 1.07^{2.8594-1} + 50 \cdot 1.07^{2.8594-2} + 1050 \cdot 1.07^{2.8594-3} \\
&= 1.07^{2.8594} \left(50 \cdot 1.07^{-1} + 50 \cdot 1.07^{-2} + 1050 \cdot 1.07^{-3}\right) \\
&\approx 1.07^{2.8594} \cdot 947.5137 \\
&\approx 1149.7554 > V(z^*, 0.05)
\end{aligned}$$

If within the first year the YTM undergoes a a variation of $\Delta i = -2\%$, the result obtained at $z^* = 2.8594$ years becomes:

$$\begin{aligned}
V(z^*, 0.03) &= 50 \cdot 1.03^{2.8594-1} + 50 \cdot 1.03^{2.8594-2} + 1050 \cdot 1.03^{2.8594-3} \\
&= 1.03^{2.8594} \left(50 \cdot 1.03^{-1} + 50 \cdot 1.03^{-2} + 1050 \cdot 1.03^{-3}\right) \\
&\approx 1.03^{2.8594} \cdot 1056.5722 \\
&\approx 1149.7567 > V(z^*, 0.05)
\end{aligned}$$

The change in value, although relatively small, is positive in both cases. No matter how the YTM varies, we take home a return that is greater than $i = 5\%$.

7.2. Bond price volatility

As we stated above, the current market price of a bond $P(i)$ depends on the yield rate i:

$$P(i) = \sum_{s=1}^{n} a_s (1+i)^{-t_s}$$

Clearly, if the yield rate undergoes a variation Δi at $\tau \in (0, t_1]$ the current price also varies:

$$\Delta P(i) = P(i + \Delta i) - P(i)$$

The *volatility* of the price of a bond is the sensitivity of the price to a change in yield rate, expressed as a percentage change:

$$\frac{\Delta P(i)}{P(i)} = \frac{P(i + \Delta i) - P(i)}{P(i)}$$

We are interested in *estimating* the percentage change of the price as a *linear function* of the possible variation Δi of the market rate.

Theorem 64. If the market price of a fixed income bond is

$$P(i) = \sum_{s=1}^{n} a_s (1+i)^{-t_s}$$

then:
(i) the relation

$$\frac{P'(i)}{P(i)} = -\frac{D(i)}{1+i}$$

holds;
(ii) a linear approximation of the *volatility* of the price is

$$\frac{\Delta P(i)}{P(i)} \approx \frac{P'(i)}{P(i)} \Delta i = -\frac{D(i)}{1+i} \Delta i$$

Proof. (i) Since

$$P'(i) = \sum_{s=1}^{n} -t_s \cdot a_s (1+i)^{-t_s-1} = -(1+i)^{-1} \sum_{s=1}^{n} t_s \cdot a_s (1+i)^{-t_s}$$

we obtain

$$\frac{P'(i)}{P(i)} = -(1+i)^{-1} \frac{\sum_{s=1}^{n} t_s \cdot a_s (1+i)^{-t_s}}{\sum_{s=1}^{n} a_s (1+i)^{-t_s}} = -\frac{D(i)}{1+i}$$

(ii) The function $P(i)$ is everywhere strictly positive and differentiable. We can consider the first order Taylor formula at every point $i \in (-1, +\infty)$:

$$P(i+\Delta i) - P(i) = P'(i) \Delta i + o(\Delta i) \text{ as } \Delta i \to 0$$

thereby

$$\frac{\Delta P(i)}{P(i)} = \frac{P(i+\Delta i) - P(i)}{P(i)} = \frac{P'(i)}{P(i)} \Delta i + \frac{o(\Delta i)}{P(i)} = \frac{P'(i)}{P(i)} \Delta i + o(\Delta i) \text{ as } \Delta i \to 0$$

By neglecting $o(\Delta i)$ we get the desired linear approximation:

$$\frac{\Delta P(i)}{P(i)} \approx \frac{P'(i)}{P(i)} \Delta i = -\frac{D(i)}{1+i} \Delta i$$

We have what we wanted: an approximation of bond price volatility as a linear function of Δi.
The ratio

$$\boxed{D^*(i) = \frac{D(i)}{1+i}}$$

is called *modified duration* and represents a measure of the volatility of the bond price:

$$\frac{\Delta P(i)}{P(i)} \approx -D^*(i) \Delta i$$

The higher it is, the greater the volatility.

Some key observations:

- In the estimation we have ignored $o(\Delta i)$. This means that the estimate is good if we consider small variations in yield rate; it is not reliable if we consider large variations.

- We confirm that the price is a decreasing function of the rate i. If $\Delta i > 0$ then $\Delta P(i) < 0$ and also $\Delta P(i)/P(i) < 0$.

- We can also derive an estimate of the absolute variation in price

$$\boxed{\Delta P(i) \approx -D^*(i)\, P(i)\, \Delta i}$$

and therefore derive an estimate of the new price

$$\boxed{P(i + \Delta i) \approx P(i) - D^*(i)\, P(i)\, \Delta i}$$

Example 65. We again consider the usual coupon bond (A) at a rate of $i = 5\%$ with price $P(0.05) = 1000$ and duration $D(0.05) = 2.8594$. The modified duration is

$$D^*(0.05) = \frac{2.8594}{1.05} \approx 2.72324$$

Considering $\Delta i = 0.8\%$ we obtain

$$\frac{\Delta P(0.05)}{P(0.05)} \approx -2.72324 \cdot 0.8\% = -2.17859\%$$

and also

$$P(0.058) - P(0.05) \approx -1000 \cdot 0.0217859 = -21.7859$$
$$P(0.058) \approx 1000 - 21.7859 = 978.2141$$

To verify the reliability of the estimate we can calculate the actual price of the bond at the changed market rate $i' = 0.058$, obtaining

$$P(0.058) = 50 \cdot 1.058^{-1} + 50 \cdot 1.058^{-2} + 1050 \cdot 1.058^{-3} = 978.5365$$

We note an error of approximately 32 cents, which on 978 euro can be considered as negligible.